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Midterm Examination - 1

Phys332: Electromagnetic Theory II

2026/03/25

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

(1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;

(2) I take *full responsibility* for any ambiguity in my selections in "multiple choice questions";

(3) incorrect selections will receive $-1/7$ of the question's points;

(4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;

(5) no partial credit is awarded for the explanations provided in the answer boxes;

(6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded negative points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;

(7) any page which does not contain *both my name and student id* may not be graded;

(8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: _____

Question:	1	2	3	4	5	Total
Points:	$31\frac{1}{2}$	$10\frac{1}{2}$	21	21	21	105

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All dimensionfull quantities are to be understood in SI units.

(For instance, $\epsilon_0 \simeq 10^{-11}$ and $|\mathbf{B}| = 5$ stand for $\epsilon_0 \simeq 10^{-11}\ \mathrm{A}^2\ \mathrm{kg}^{-1}\ \mathrm{m}^{-3}\ \mathrm{s}^4$ and $|\mathbf{B}| = 5\ \mathrm{A}^{-1}\ \mathrm{kg}^1\ \mathrm{s}^{-2}$.)

Question: 1: Eddy currents ..... *(31½ points)*

(Modified version of Example 7.5 of 978-1-009-39775-9, which we partially solved in class)

For a rigid rotating object of angular velocity $\boldsymbol{\omega}$, linear velocity is given as $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$.
Please take $\epsilon_0 \simeq 10^{-11}$ and $\mu_0 \simeq 10^{-6}$ for this question.
One can derive from the Maxwell equations (see the formulae sheet) the continuity equation $\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$.
Please take Ohm's law as $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

Imagine a uniform, electrically neutral, spherical shell, of inner radius R_1, outer radius R_2, and conductivity σ, centered at the origin and spinning with constant angular velocity $\boldsymbol{\omega}$, in the presence of a uniform magnetic field $\mathbf{B}$. We will compute the eddy currents in the shell, in the steady state (fields, charges, currents all constant in time).

(a) **(10½ points)** Which one below would give the right expression for charge density ρ? *Hint: combine Ohm's law, Gauss law, and continuity equation.*

☐ $\rho = \epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ■ $\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \mu_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = -\mu_0 \nabla \cdot (\mathbf{v} \times \mathbf{B})$

☐ $\rho = \epsilon_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = -\epsilon_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = \mu_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$ ☐ $\rho = -\mu_0^{-1} \nabla \cdot (\mathbf{v} \times \mathbf{B})$

(b) **(10½ points)** For $\mathbf{B} = 5\hat{z}$ and $\boldsymbol{\omega} = 10\hat{z}$, what would be ρ?

■ $\rho = -10^{-9}$ ☐ $\rho = 10^{-9}$ ☐ $\rho = -5 \times 10^{-10}$ ☐ $\rho = 5 \times 10^{-10}$

☐ $\rho = -10^{13}$ ☐ $\rho = 10^{13}$ ☐ $\rho = -5 \times 10^{12}$ ☐ $\rho = 5 \times 10^{12}$

(c) **(10½ points)** Consider two points, point A on the outer surface and point B on the inner surface, and assume that the origin and these two points form a line. What would be the ratio $\left(\frac{\partial V}{\partial r}\Big|_{\text{at the point } A}\right) / \left(\frac{\partial V}{\partial r}\Big|_{\text{at the point } B}\right)$ for the electric potential V and the radial distance r? *Hint: combine Ohm's law along with the fact that eddy currents here have no radial components.*

☐ $-\sqrt{R_2/R_1}$ ☐ $-\sqrt{R_1/R_2}$ ☐ $\sqrt{R_2/R_1}$ ☐ $\sqrt{R_1/R_2}$ ☐ $-R_2/R_1$ ☐ $-R_1/R_2$ ■ R_2/R_1 ☐ R_1/R_2

Solution 1.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), as stated in the hint, we can combine continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = 0 \text{ (since in steady state)} \tag{1}$$

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with Ohm's law, $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$*, to get*

$$\nabla \cdot \mathbf{E} = -\nabla \cdot (\mathbf{v} \times \mathbf{B}) \tag{2}$$

which gives the required expression via Gauss law:

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) \tag{3}$$

For part (b), we simply insert the relevant quantities:

$$\mathbf{v} \times \mathbf{B} = (\boldsymbol{\omega} \times \mathbf{r}) \times \mathbf{B} = \mathbf{r}(\mathbf{B} \cdot \boldsymbol{\omega}) - \boldsymbol{\omega}(\mathbf{B} \cdot \mathbf{r}) = |\boldsymbol{\omega}||\mathbf{B}|(x\hat{x} + y\hat{y}) \tag{4}$$

hence

$$\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) = -2\epsilon_0 |\boldsymbol{\omega}||\mathbf{B}| \simeq -2 \times 10^{-11} \times 10 \times 5 = -10^{-9} \tag{5}$$

For part (c), use the hint:

$$\mathbf{J} \cdot \hat{r} = \sigma\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \hat{r} = 0 \quad \rightarrow \quad \mathbf{E} \cdot \hat{r} = -(\mathbf{v} \times \mathbf{B}) \cdot \hat{r} \tag{6}$$

hence

$$\frac{\partial V}{\partial r} = \hat{r} \cdot \nabla V = -\hat{r} \cdot \mathbf{E} = (\mathbf{v} \times \mathbf{B}) \cdot \hat{r} = [(\boldsymbol{\omega} \times \mathbf{r}) \times \mathbf{B}] \cdot \hat{r} \tag{7}$$

Since magnetic field is uniform and the angular velocity is constant, the only difference at the points A and B will be due to $\mathbf{r}$*. But they are collinear, so we simply have* $\mathbf{r}\big|_A = R_2\hat{s}$ *and* $\mathbf{r}\big|_B = R_1\hat{s}$ *for some unit vector* $\hat{s}$ *if the origin is not part of the* $|AB|$ *line segment (*$\mathbf{r}\big|_A = R_2\hat{s}$ *and* $\mathbf{r}\big|_B = R_1(-\hat{s})$ *if the origin is part of the* $|AB|$ *line segment). Thus:*

$$\left(\frac{\partial V}{\partial r}\bigg|_{\text{at the point } A}\right) \Big/ \left(\frac{\partial V}{\partial r}\bigg|_{\text{at the point } B}\right) = \pm\frac{R_2}{R_1} \tag{8}$$

Since the question does not specify the case, both of these answers are taken to be correct.

Question: 2: Mutual inductance ..... *(10½ points)*
(Modified version of Example 7.11 of 978-1-009-39775-9)

Please take $\pi\mu_0 \simeq 4 \times 10^{-6}$ *for this question.*

A cylindrical solenoid (length 3 and radius 4, with 75 turns per unit length) lies coaxially inside an infinitely long cylindrical solenoid (radius 5, with 500 turns per unit length). Current $I = 5$ flows in the finite solenoid. What is the flux through the infinite solenoid?
Hint: Mutual inductance is a symmetric object, i.e. $M_{12} = M_{21}$ *for* $\Phi_1 = M_{12}I_2$ *and* $\Phi_2 = M_{21}I_1$.

☐ 24 ☐ 32 ■ 36 ☐ 48 ☐ 60 ☐ 72 ☐ 96 ☐ 108

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Solution 2.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

The total flux through the infinite solenoid is hard to compute as the flux generated by a finite size solenoid is computationally cumbersome. Luckily (and as hinted in the question), we can solve a different question where current is in the infinite solenoid and we compute the flux through the finite solenoid: since mutual inductance is same, the answer to this new question will be exactly same to the original question!

The magnetic field generated by an infinite solenoid is straightforward via Faraday's law:

$$B = \mu_0 n I = \frac{4}{\pi} \times 10^{-6} \times 500 \times 5 = \frac{1}{\pi} \times 10^{-2} \tag{9}$$

The flux through a single loop of the finite size solenoid then is

$$\Phi_{per\ single\ turn} = B\pi r^2 = \frac{1}{\pi} \times 10^{-2} \times \pi \times 4^2 = 16 \times 10^{-2} \tag{10}$$

As there are 3×75 *turns in total, the total flux (hence the answer) reads*

$$\Phi = 16 \times 10^{-2} \times 3 \times 75 = 36 \tag{11}$$

Question: 3: Poynting vector ..... *(21 points)*
(Modified version of Example 8.1 of 978-1-009-39775-9, which we partially solved in class)

Please take $\epsilon_0 \simeq 10^{-11}$ *and* $\mu_0 \simeq 10^{-6}$ *for this question.*

Imagine a cylindrical conductive wire of radius $\pi^{-1/2}$ and length 2 in an otherwise empty space. Set up a cylindrical coordinate system such that the axis of the wire coincides with the positive z–axis, with one of its faces coinciding with the $z = 0$ plane. We will assume that the $z = 0$ face of the wire is held at the electric potential $V = 9$ and the other face at $V = 3$. We are additionally told that the current density $\mathbf{J}$ satisfies $\mathbf{J}(z = 0, r, \theta) = 2\hat{z}$.

(a) **(10½ points)** Assuming that the wire has a constant resistivity, what is the total work done per unit time by the electric & magnetic fields on the material?

☐ 5 ☐ 6 ☐ 7 ■ 12 ☐ −5 ☐ −6 ☐ −7 ☐ −12

(b) **(10½ points)** Assuming that the wire has a conductivity inversely linear in z such that $\sigma(z)^{-1} = 1 + cz$ for some constant c (to be determined by the given electric potential and the current density), what is the total work done per unit time by the electric & magnetic fields on the *part* of the material between $z = 0$ and $z = 1$?
Hint: resistance is the volume integral of resistivity and resistivity is the inverse of conductivity.

■ 5 ☐ 6 ☐ 7 ☐ 12 ☐ −5 ☐ −6 ☐ −7 ☐ −12

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Solution 3.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

By integrating the current density at $z = 0$, we can find the current at $z = 0$; but since there is no way for current to leak to some other system (as there is none), total current will be constant throughout the wire:

$$I = \int_{z=0 \text{ cross-section}} \mathbf{J} \cdot \mathbf{dA} = |\mathbf{J}| \int_{z=0 \text{ cross-section}} |\mathbf{dA}| = 2 \times \pi \times (\pi^{-1/2})^2 = 2 \tag{12}$$

From Faraday's law, this then leads to a constant magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\theta} = \frac{10^{-6} \times 2}{2 \times \pi \times \pi^{-1/2}}\hat{\theta} = \frac{10^{-6}}{\pi^{1/2}}\hat{\theta} \tag{13}$$

In part (a), the potential should change linearly due to the constant resistivity (and the cylindrical symmetry in the setup): with given boundary conditions ($V(0) = 9$ and $V(2) = 3$), we conclude that $V(z) = 9-3z$ which indicates

$$\mathbf{E} = -\nabla V = 3\hat{z} \tag{14}$$

The Poynting vector is then

$$\mathbf{S} = \frac{1}{\mu}\mathbf{E} \times \mathbf{B} = 10^6(3\hat{z}) \times \left(\frac{10^{-6}}{\pi^{1/2}}\hat{\theta}\right) = -\frac{3}{\pi^{1/2}}\hat{r} \tag{15}$$

*To compute the total work done per unit time **by** the fields **on** the material, we should integrate this vector against the surface pointing inward towards the material, hence*

$$\begin{aligned}\int \mathbf{S} \cdot \mathbf{dA} &= \int_{z=0 \text{ face}} \mathbf{S} \cdot \hat{z}|\mathbf{dA}| + \int_{z=2 \text{ face}} \mathbf{S} \cdot (-\hat{z})|\mathbf{dA}| + \int_{\text{side face}} \mathbf{S} \cdot (-\hat{r})|\mathbf{dA}| \\ &= 0 + 0 + \frac{3}{\pi^{1/2}} \int_{\text{side face}} |\mathbf{dA}| = \frac{3}{\pi^{1/2}}(2 \times \pi \times \pi^{-1/2} \times 2) = 12\end{aligned} \tag{16}$$

As expected, the answer is simply $\Delta V \times I = (9 - 3) \times 2 = 12$.

For part (b), we can follow a similar approach: find the functional form of V in z (it will not be linear anymore), find E, find S, and integrate S. Instead, it may be shorter to find resistances of the two parts of the material, $z \in (0, 1)$ part and $z \in (1, 2)$ part, use the fact that they are actually in series, and

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then relate them to the total expected resistance $\Delta V/I$. The relevant resistances are

$$\begin{aligned} R_1 =& (\pi \times (\pi^{-1/2})^2)^{-1} \int_0^1 (1+cz)dz = \frac{c+2}{2} \\ R_2 =& (\pi \times (\pi^{-1/2})^2)^{-1} \int_1^2 (1+cz)dz = \frac{3c+2}{2} \end{aligned} \tag{17}$$

which follow from the infinitesimal version of the freshman law $R = \rho\frac{\ell}{A}$, i.e. $dR = \rho\frac{d\ell}{A}$. Unfortunately, the hint is incorrect for this relation (resistance is not *volume integral of resistivity); however, for this particular question, hint does not do any harm as the area is* 1 *(hence one gets the correct final answer even if they use the hint).*

As the potential difference and the total current are still same (given as input in the question), we can now find c:

$$R_1 + R_2 = \frac{\Delta V}{I} \quad \Rightarrow \quad 2(c+1) = \frac{9-3}{2} \quad \Rightarrow \quad c = \frac{1}{2} \quad \Rightarrow \quad R_1 = \frac{5}{4} \tag{18}$$

We can now straightforwardly write down the total work done per unit time by the electric & magnetic fields on the part of the material between $z = 0$ and $z = 1$ as

$$\frac{dW}{dt} = P = RI^2 = \frac{5}{4}2^2 = 5 \tag{19}$$

Question: 4: Classical Waves ..... *(21 points)*

Wave equation reads as $\left(\nabla^2 - \frac{1}{v^2}\frac{\partial^2}{\partial t^2}\right) f(\mathbf{r}, t) = 0$ *for the laplacian* ∇^2.

Any function f that satisfies wave equation is called a classical wave.

(a) **(10½ points)** Which $f(x, y, z, t)$ below describes a classical wave, of speed $\sqrt{2}$, that propagates along the positive $\hat{x} + \hat{y}$ direction?

■ $\frac{1}{2x+2y-4t}$ ☐ $\tan(2x+2y+4t)$ ☐ $\cos(4x+4y+2t)$ ☐ $e^{4x+4y-2t}$

☐ $e^{i(4x+4y+2\sqrt{2}t)}$ ☐ $4x+4y-2\sqrt{2}t$ ☐ $-2x-2y+2\sqrt{2}t$ ☐ $-\frac{1}{2x+2y+2\sqrt{2}t}$

(b) **(10½ points)** Consider a one dimensional classical wave

$$f(x,t) = (a+3i)\exp(i(bx-vt)) + (c+id)\exp(i(x+vt)) \tag{20}$$

For which values of (a, b, c, d) would this function describe a real plane wave solution?

☐ $(1,-1,-2,-3)$ ☐ $(2,-1,-2,-3)$ ☐ $(-1,-1,-2,-3)$ ■ $(-2,-1,-2,-3)$

☐ $(1,-1,-2,3)$ ☐ $(2,-1,-2,3)$ ☐ $(-1,-1,-2,3)$ ☐ $(-2,-1,-2,3)$

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Solution 4.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), we first observe that $f(x, y, z, t) = g(\hat{n}\cdot\mathbf{r} - vt)$ (for any function g) describes a wave that propagates along positive $\hat{n}$ direction with the speed v. Therefore, we need $f(x, y, z, t) = g\left(\frac{\hat{x}+\hat{y}}{\sqrt{2}}\cdot\mathbf{r} - \sqrt{2}t\right)$ which is only satisfied by the first option for the function $g(\alpha) = 1/(2\sqrt{2}\alpha)$.

For part (b), we simply use the fact that a real function f satisfies $f^\dagger = f$, thus

$$\left[(a+3i)e^{i(bx-vt)} + (c+id)e^{i(x+vt)}\right]^\dagger = (a+3i)e^{i(bx-vt)} + (c+id)e^{i(x+vt)} \tag{21}$$

hence, for real unknowns, we get

$$(c-id)e^{i(-x-vt)} + (a-3i)e^{i(-bx+vt)} = (a+3i)e^{i(bx-vt)} + (c+d)e^{i(x+vt)} \tag{22}$$

which requires $a = c$, $d = -3$, and $b = -1$.

Question: 5: Concept of polarization . . . . . *(21 points)*

(a) **(10½ points)** We know that in free space the Gauss law dictates $\nabla\cdot\mathbf{E} = 0$, which in turn means that the $\mathbf{E}$ field does not have a *longitudinal polarization* component. If we are given

$$\mathbf{E}(x, y, z, t) = 3(\hat{x}+\hat{y}+\hat{z})\cos(\alpha x + \beta y + \gamma z - ct + \delta) \tag{23}$$

Which option below for the values of $(\alpha, \beta, \gamma, \delta)$ gives a solution for the electrical field that is consistent with the Gauss law?

☐ $(1, 1, 1, -1)$ ☐ $(1, 1, 1, 0)$ ☐ $(1, 1, 1, 1)$ ☐ $(-1, -1, -1, -1)$

☐ $(-1, -1, -1, 0)$ ☐ $(-1, -1, -1, 1)$ ☐ $(1, 1, 2, 0)$ ■ $(1, 1, -2, 0)$

(b) **(10½ points)** Consider the hypothetical situation where Gauss law gets modified in free space to the form

$$\nabla\cdot\mathbf{E} = -\frac{4}{c}\frac{\partial(\hat{x}\cdot\mathbf{E})}{\partial t} \tag{24}$$

Which option below for the values of $(\alpha, \beta, \gamma, \delta)$ gives a solution for electrical field consistent with this modified version of the Gauss law?

☐ $(1, 1, 1, -1)$ ☐ $(1, 1, 1, 0)$ ☐ $(1, 1, 1, 1)$ ☐ $(-1, -1, -1, -1)$

☐ $(-1, -1, -1, 0)$ ☐ $(-1, -1, -1, 1)$ ■ $(1, 1, 2, 0)$ ☐ $(1, 1, -2, 0)$

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Solution 5.1 - Below is the full derivation of the result; in the exam, you are only supposed to describe the procedure without any actual derivation in the answer box, and do so in a succinct, coherent and well-articulated manner.

For part (a), we simply need to impose $\nabla \cdot \mathbf{E} = 0$*, which gives* $\alpha + \beta + \gamma = 0$*. The only solution that satisfies this is the last one.*

For part (b), we instead impose

$$\begin{aligned} 0 =& \nabla \cdot \mathbf{E} + \frac{4}{c}\frac{\partial(\hat{x} \cdot \mathbf{E})}{\partial t} \\ 0 =& [3(\alpha + \beta + \gamma)\sin(\alpha x + \beta y + \gamma z - ct + \delta)] + \frac{4}{c}[-3c\sin(\alpha x + \beta y + \gamma z - ct + \delta)] \\ 0 =& 3(\alpha + \beta + \gamma - 4)\sin(\alpha x + \beta y + \gamma z - ct + \delta) \end{aligned} \tag{25}$$

which is only zero if $\alpha + \beta + \gamma - 4 = 0$ *and that is only satisfied by the seventh option.*

It is interesting to note that this modified version of Gauss law allows longitudinal polarization for the electric field, but we now have only one transverse polarization: the total number of polarization is still two, as we still impose a condition due to the (modified or not) Gauss law.

« « « Congratulations, you have made it to the end! » » »